Day_07_IPConfiguration

THEORY (8 marks)

1. What is a network address?
   A network address is a unique identifier used to label a specific network connected to computer network.

2. What is a broadcast address?
   A network address that is special used to send data to every network simultaneously connected through a local network segment.

3. What is a usable host address?
   A usable host address is an IP address that can be assigned to a network such as server, printer, computers and etc…

4. What is the first usable IP address?
   The first usable IP address is the address immediately following the network address.

5. What is the last usable IP address?
   The last broadcast IP address minus 1 is the Last usable IP address.

6. What does /24 mean Class C

7. What subnet mask corresponds to /24: 255.255.255.0

8. What is a default gateway?
   Default gateway is where router sends traffic outside the local network or segment.

9. Why does a router interface need an IP address?
   A router interface needs an IP address because it can belong to specific network or subnet, acts as the default gateway for devices on the network and forward data between packet between different networks.

10. What is a static IP configuration?
    Static IP configuration is a permanent, unchanged numerical label assigned to device on a network.

11. What does "show ip interface brief" display?
    It displays summary of interface, their IP addresses and their status.

12. What does "no shutdown" do conceptually on a Cisco interface?
    It enables the interface making it become operational.

13. What does interface status "up/up" generally indicate?
    Both physical interface and data link are operational.

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14. Why should network configuration always be verified?
    To make sure configuration is standard, network works efficiently and errors are rectified.

15. Why is IPv4 configuration important in cybersecurity?
    It helps find and manage device communication also detects suspicious activities.

IPv4 EXERCISE (4 marks)

For 192.168.100.0/24 identify:
- Network Address: 192.168.100.0
- First Usable IP: 192.168.100.1
- Last Usable IP: 192.168.100.254
- Broadcast Address: 192.168.100.255

PRACTICAL (4 marks)

From YOUR OWN PC (ipconfig /all) record:

IPv4 Address: 10.101.139.72

Subnet Mask: 255.255.224.0

Default Gateway: 10.101.128.1

DNS Server: 9.9.9.9, 208.67.222.222, 8.8.8.8

SCREENSHOTS (2 marks)

Attach screenshots of: ipconfig /all · ping 127.0.0.1 · ping <your default gateway> · Packet Tracer topology (if you completed Part 4).

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C:\Windows\System32>ipconfig/all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Kirthikaa
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-05
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::1b54:823e:5810:faec%5(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 621412391
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : 26-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 2A-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
```

```
Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : 2A-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter VMware Network Adapter VMnet1:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet1
   Physical Address. . . . . . . . . : 00-50-56-C0-00-01
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::7a44:822:bc1f:b151%15(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.150.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 12 October 2026 9.08.30 AM
   Lease Expires . . . . . . . . . . : 12 October 2026 3.31.06 PM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.150.254
   DHCPv6 IAID . . . . . . . . . . . : 201347158
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter VMware Network Adapter VMnet8:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VMware Virtual Ethernet Adapter for VMnet8
   Physical Address. . . . . . . . . : 00-50-56-C0-00-08
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::680e:e4a8:74cf:3ece%2(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.35.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : 12 October 2026 9.08.30 AM
   Lease Expires . . . . . . . . . . : 12 October 2026 3.31.07 PM
   Default Gateway . . . . . . . . . :
   DHCP Server . . . . . . . . . . . : 192.168.35.254
   DHCPv6 IAID . . . . . . . . . . . : 234901590
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   Primary WINS Server . . . . . . . : 192.168.35.2
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   Primary WINS Server . . . . . . . : 192.168.35.2
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek RTL8852BE WiFi 6 802.11ax PCIe Adapter
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-75
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::450d:aba6:dee6:4cd7%18(Preferred)
   IPv4 Address. . . . . . . . . . . : 10.101.139.72(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.224.0
   Lease Obtained. . . . . . . . . . : 12 October 2026 9.08.44 AM
   Lease Expires . . . . . . . . . . : 12 October 2026 6.15.37 PM
   Default Gateway . . . . . . . . . : 10.101.128.1
   DHCP Server . . . . . . . . . . . : 10.101.128.1
   DHCPv6 IAID . . . . . . . . . . . : 153399993
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-30-09-D0-3E-00-EE-BC-DB-4B-B3
   DNS Servers . . . . . . . . . . . : 9.9.9.9
                                       208.67.222.222
                                       8.8.8.8
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 24-B2-B9-9F-EC-76
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

```
C:\Windows\System32>ping 127.0.0.1

Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Windows\System32>ping 10.101.128.1

Pinging 10.101.128.1 with 32 bytes of data:
Reply from 10.101.128.1: bytes=32 time=18ms TTL=254
Reply from 10.101.128.1: bytes=32 time=4ms TTL=254
Reply from 10.101.128.1: bytes=32 time=10ms TTL=254
Reply from 10.101.128.1: bytes=32 time=5ms TTL=254

Ping statistics for 10.101.128.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 4ms, Maximum = 18ms, Average = 9ms

C:\Windows\System32>
```